



Modelling Financial Markets

Seminar 13: Machine Learning in Finance — Hands-on



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Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ CI: Confidence Interval
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ EV: Explained Variation
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FFD: Fixed-width window Fractional Differentiation
- ▣ FFN: FeedForward Network
- ▣ FN: False Negative
- ▣ GAN: Generative Adversarial Network
- ▣ GB: Gradient Boosting
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LS: Linear SDF estimated by unregularised Least Squares
- ▣ LW: Local Whittle (estimator)
- ▣ MA: Moving Average

Acronyms used in this material (2/2)

- ▣ MA20: 20-day Moving Average
- ▣ MDA: Mean Decrease Accuracy
- ▣ MDI: Mean Decrease Impurity
- ▣ ML: Machine Learning
- ▣ OLS: Ordinary Least Squares
- ▣ PBO: Probability of Backtest Overfitting
- ▣ PSR: Probabilistic Sharpe Ratio
- ▣ PT: Pesaran–Timmermann (test of directional accuracy)
- ▣ RC: Reality Check (White's test for data snooping)
- ▣ RF: Random Forest
- ▣ ROC: Receiver Operating Characteristic (curve)
- ▣ RSI: Relative Strength Index
- ▣ SD: Standard Deviation
- ▣ SDF: Stochastic Discount Factor
- ▣ SE: Standard Error
- ▣ SHAP: SHapley Additive exPlanations
- ▣ SPA: Superior Predictive Ability (test)
- ▣ SR: Sharpe Ratio
- ▣ TN: True Negative
- ▣ TP: True Positive
- ▣ US: United States
- ▣ VIX: CBOE Volatility IndeX

Outline

Objectives and Setup

Part A: Pen-and-Paper Problems

Part B: Python Problems on Crypto Data

Part C: Open Problem and AI Critique

Assignment and Further Study

Key Definitions (1/4): Data, Labels, Validation

Features and labels

- **FFD**: $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$, weights truncated when $|w_k| < \tau$
- **ADF (Dickey & Fuller (1979))**: H_0 : unit root; reject if the statistic is below the 5% critical value (a rejection is not proof of stationarity: A2)
- **EWMA**: standard deviation of daily log returns with exponentially decaying weights (span 50)
- **Triple barrier**: label $+1/-1$ if the upper/lower barrier (pt , $s/\text{multiples of } \sigma_{t_0} \sqrt{h}$) is hit first; else the sign at $t_0 + h$

Validation

- **K-fold CV**: K blocks; fit on $K - 1$, test on 1, repeat
- **Leakage**: training data contain information that is also in the test data
- **Purging / embargo**: drop training observations whose label span overlaps the test block, and a few just after it

Key Definitions (2/4): Models and Classification Metrics

Models and signals

- ▣ **Logit-L1**: logistic regression with an L_1 (LASSO) penalty
- ▣ **RF**: average of decorrelated trees; **GB**: small trees added sequentially
- ▣ **MA crossover**: long when the fast MA is above the slow one

Classification metrics

- ▣ Accuracy = $\frac{TP+TN}{n}$; precision = $\frac{TP}{TP+FP}$; recall = $\frac{TP}{TP+FN}$; $F_1 = \frac{2PR}{P+R}$ (TP, FP, TN, FN: true/false positives/negatives)
- ▣ **ROC AUC**: $P(\hat{s}_+ > \hat{s}_-) + \frac{1}{2}P(\hat{s}_+ = \hat{s}_-)$ for a random positive and a random negative; 0.5 = no skill
- ▣ **Log loss**: $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$, the loss of a probability forecast



Key Definitions (3/4): Inference and Backtests

Uncertainty

- ▣ **95% CI**: estimate ± 1.96 SE
- ▣ **Block bootstrap**: resample blocks of L consecutive observations to keep serial dependence (Künsch (1989))
- ▣ **HAC**: Newey–West variance (Newey & West (1987)), valid under serial correlation
- ▣ **DM**: Diebold–Mariano test of equal predictive accuracy: t -test on the loss differential $d_t = \ell_t^A - \ell_t^B$ with HAC SE (Diebold & Mariano (1995))

Backtests

- ▣ **SR**: $\bar{r}/\hat{\sigma}$ of daily log returns; annualised $\times \sqrt{365}$ for crypto
- ▣ **PSR / DSR** (Bailey & López de Prado (2012); Bailey & López de Prado (2014)): use skewness γ_3 , kurtosis γ_4 (Normal distribution: 3)
- ▣ N_{eff} : effective number of independent trials
- ▣ **PBO**: estimated by CSCV (Bailey, Borwein, López de Prado & Zhu (2017))

How This Seminar Works

- ▣ Two kinds of problems, in every part
 - ▶ **[Solved]**: worked models, with full solutions in this deck and in the notebook
 - ▶ **[Proposed]**: you solve them; each follows an analogous solved problem
- ▣ Part A: pen and paper (derivations, small numerical cases)
- ▣ Part B: Python on Bitcoin, verification *and* inference: every number with its uncertainty
 - ▶ each exercise ends with interpretation questions: what would change the conclusion?
- ▣ Part C: open questions that can seed your team project (C1, C3)
 - ▶ and an AI answer to critique (C2): find the leaks, correct them, log them in `AI_ERRORS.md`
- ▣ Rules: no look-ahead, purged validation, a naive benchmark, and a log of every trial ([Harvey, Liu & Zhu \(2016\)](#))

Key Definitions (4/4): Memory and Predictive Ability

Memory

- $I(\delta)$: $(1 - B)^\delta X_t$ is $I(0)$ (stationary, spectral density finite and positive at frequency 0); X_t itself is stationary only for $\delta < \frac{1}{2}$
- **LW / ELW**: local Whittle (Robinson (1995)) and exact local Whittle (Shimotsu & Phillips (2005)) estimators of δ from the first m Fourier frequencies; $SE = 1/(2\sqrt{m})$

Predictive ability

- **PT**: Pesaran–Timmermann test of sign independence (Pesaran & Timmermann (1992))
- **CW**: Clark–West test for nested models (Clark & West (2007)); **GW**: Giacomini–White test of conditional predictive ability (Giacomini & White (2006))
- **RC / SPA / StepM**: bootstrap tests of the best of many rules (White (2000); Hansen (2005); Romano & Wolf (2005))

Problem Overview

	[Solved]	[Proposed]
A	A1 FFD weight sums; A3 barrier hitting probabilities; A5 SR variance, PSR, DSR; A7 cost of purging; A9 AUC and its SE	A2 ADF over-rejection; A4 barriers with drift; A6 DSR and the Gumbel constant; A8 purging; A10 DeLong variance
B	B1 BTC memory; B3 triple barrier; B5 purged CV; B7 models and AUC CI; B9 meta-labelling; B10 DSR and effective N ; B12 PT and CW	B2 ETH memory; B4 asymmetric barriers; B6 find the leak; B8 Diebold–Mariano; B11 PBO; B13 SPA vs. DSR; B14 double selection
C		C1 foundation model vs. boosting; C2 critique an AI answer; C3 no-arbitrage SDF on decile portfolios

- Each proposed problem names its model: the analogous solved problem

Setup: Python Environment and Data

Environment

- Google Colab (recommended) or local Python ≥ 3.10
- numpy, pandas, scipy, statsmodels, scikit-learn, shap
-  **Open in Colab:**
chapter13_seminar_notebook.ipynb

```
import pandas as pd
url = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/BTC-USD.CC.csv'
btc = pd.read_csv(url, parse_dates=['date']).set_index('date')['close'].loc['2014-09-17':]
```

Data

- **Bitcoin in dollars (BTC-USD)**, daily, 2014–2026 (4 385 obs.)
- **Ether (ETH-USD)**, daily, 2015–2026 (4 061 obs.)
- Files in the course repository: data/market

A1 [Solved]: FFD Weights, Their Sums and the Window Length

Task

- **Context:** fractional differentiation of order d with weights $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k}$, truncated at the first $|w_k| < \tau$ (Hosking (1981); López de Prado (2018)); no data are needed
- **Tasks:**
 - ▶ (a) Show that $w_k = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)}$ and that $w_k \sim k^{-(1+d)}/\Gamma(-d)$ for large k .
 - ▶ (b) Show that $\sum_{k=0}^K w_k = \frac{\Gamma(K+1-d)}{\Gamma(1-d)\Gamma(K+1)} \sim K^{-d}/\Gamma(1-d)$.
 - ▶ (c) For $d = 0.25$ and $\tau = 10^{-4}$ compute the window K and $\sum_{k \leq K} w_k$.
 - ▶ (d) Decide whether the FFD series of an $I(1)$ log price is stationary.
- **Report:** the two formulas, K , the weight sum and the decision in (d)

Interpretation

- Why do the lecture and the literature call the FFD series “stationary”, and what is actually stationary in it?

A1: Solution (1/2)

Solution

- (a) w_k is the coefficient of B^k in $(1 - B)^d = \sum_k \binom{d}{k} (-B)^k$, and $(-1)^k \binom{d}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)}$
 - Stirling: $\Gamma(k-d)/\Gamma(k+1) \sim k^{-d-1}$, so $w_k \sim k^{-(1+d)}/\Gamma(-d)$: a hyperbolic, not exponential, decay
- (b) The partial sums of the coefficients of a series $f(B)$ are the coefficients of $f(B)/(1 - B)$; here $(1 - B)^d/(1 - B) = (1 - B)^{d-1}$
 - so $\sum_{k \leq K} w_k = \frac{\Gamma(K+1-d)}{\Gamma(1-d)\Gamma(K+1)} \sim K^{-d}/\Gamma(1-d)$, which tends to 0 only as $K \rightarrow \infty$



A1: Solution (2/2)

Solution

- (c) Set $|w_K| = \tau$: $K \approx (\tau |\Gamma(-d)|)^{-1/(1+d)} = (10^{-4} \cdot 4.902)^{-0.8} = 444$ lags (the recursion also stops at 444)
- Weight sum: $K^{-d}/\Gamma(1-d) = 444^{-0.25}/1.2254 = 0.2177/1.2254 = 0.178$
- (d) Decompose: $\tilde{X}_t = (\sum_{k \leq K} w_k) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$
 - ▶ if X_t is $I(1)$, every $X_t - X_{t-k}$ is $I(0)$, so the second term is stationary
 - ▶ the first term is $0.178 X_t$: a scaled random walk, hence $\tilde{X}_t$ is still $I(1)$
- Reading: only the combination $X_t - X_{t-k}$ is stationary; the “memory” kept by FFD is a small multiple of the price level itself

A2 [Proposed]: Window Length, Weight Sums and ADF Over-Rejection

Task

- **Context:** the formulas of A1 with $\tau = 10^{-4}$; a Monte Carlo on pure random walks $X_t = \sum_{s \leq t} \varepsilon_s$, $\varepsilon_s \sim N(0, 1)$ i.i.d., seed 42
- **Tasks:**
 - ▶ (a) For $d = 0.3$ and $d = 0.5$ compute $K(\tau)$ and $\sum_{k \leq K} w_k$ with the closed forms of A1.
 - ▶ (b) Simulate 500 random walks, apply FFD with each d and keep $T = 2,500$ FFD values per path.
 - ▶ (c) Run the ADF test (constant, 1 lag) on each FFD path and record how often it rejects at 5%.
- **Report:** a table with d , K , the weight sum and the rejection frequency
- **Model:** A1

Interpretation

- The nominal significance level is 5%: why does the ADF confuse a scaled random walk with a stationary series? Which d would make an $I(1)$ price stationary?

A3 [Solved]: Triple-Barrier Labels and Hitting Probabilities

Task

- **Context:** entry price $P_{t_0} = 100$, daily volatility $\sigma_{t_0} = 1\%$, vertical barrier $h = 5$ days, $pt = sl = 1$, horizontal barriers $P_{t_0} e^{\pm \sigma_{t_0} \sqrt{h}}$ (López de Prado (2018)); for (b)–(c) the log price is σW_t , a driftless Brownian motion
- **Tasks:**
 - ▶ (a) Label path 1 (100, 101.0, 100.4, 102.5, 101.2, 99.8) and path 2 (100, 99.2, 98.6, 99.5, 98.9, 99.1).
 - ▶ (b) With barriers $+a = pt \sigma \sqrt{h}$ and $-b = -sl \sigma \sqrt{h}$ and no vertical barrier, derive $P(\text{upper first})$ and $\mathbb{E}[\tau]$ by optional stopping.
 - ▶ (c) For $pt = sl = 1$ compute the probability of reaching the vertical barrier, with continuous and with daily monitoring ($h = 10$).
- **Report:** the two labels, the two formulas and the two time-out probabilities

Interpretation

- The lecture finds 59.0% time-outs on the S&P 500 and B3 finds 59.0% on Bitcoin: what explains the gap to the Brownian value?

A3: Solution (1/2)

Solution

- (a) Width $0.01\sqrt{5} = 0.02236$: upper $100e^{0.02236} = 102.26$, lower 97.79
 - ▶ path 1: day 3, $102.5 \geq 102.26 \Rightarrow$ label +1 (the later fall does not matter)
 - ▶ path 2: no horizontal touch; $\ln(99.1/100) = -0.90\%$ at day 5 $\Rightarrow$ label -1
- (b) W_t is a martingale: $0 = p a - (1 - p) b \Rightarrow p = \frac{b}{a+b} = \frac{sl}{pt+sl}$
- $\sigma^2 W_t^2 - \sigma^2 t$ is a martingale: $\sigma^2 \mathbb{E}[\tau] = p a^2 + (1 - p) b^2 = ab \Rightarrow \mathbb{E}[\tau] = ab/\sigma^2 = pt \cdot sl \cdot h \text{ days (Karatzas \& Shreve (1998))}$

A3: Solution (2/2)

Solution

- (c) Exit time of $|W_s| < 1$ on $[0, 1]$ (series solution of the heat equation):
 - $P(\tau > h) = \frac{4}{\pi} \sum_{n \geq 0} \frac{(-1)^n}{2n+1} e^{-(2n+1)^2 \pi^2 / 8} = 1.2732 \cdot 0.2912 - \dots = 0.371$
- Daily monitoring misses touches between closes: shift each barrier out by 0.5826σ (Broadie, Glasserman & Kou (1997)), i.e. $c = 1 + 0.5826/\sqrt{10} = 1.184$; replacing $\pi^2/8$ by $\pi^2/(8c^2)$ gives 0.528
- Reading: with this approximate continuity correction, discrete monitoring explains most of the gap from 37% to 59%; the rest may reflect fat tails, volatility clustering and error in the EWMA volatility (which lags turbulent periods); this comparison does not separate them

A4 [Proposed]: Asymmetric Barriers with Drift

Task

- **Context:** warm-up with $P_{t_0} = 50$, $\sigma_{t_0} = 2\%$, $h = 4$, $pt = 2$, $sl = 1$; then the log price $X_t = \mu t + \sigma W_t$ with barriers $+a = pt \sigma \sqrt{h}$ and $-b = -sl \sigma \sqrt{h}$
- **Tasks:**
 - ▶ (a) Compute both barriers and label the paths (i) 50, 51.5, 53.0, 49.5, 47.9 and (ii) 50, 52.0, 54.5, 53.0, 51.0.
 - ▶ (b) Using the scale function $s(x) = e^{-2\mu x / \sigma^2}$, derive $P(\text{upper first}) = \frac{e^{2\mu b / \sigma^2} - 1}{e^{2\mu b / \sigma^2} - e^{-2\mu a / \sigma^2}}$ and show that it tends to $\frac{b}{a+b}$ as $\mu \rightarrow 0$.
 - ▶ (c) Evaluate the probability for $\sigma = 3\%$, $\mu = 0.1\%$ per day, $h = 10$, $pt = 2$, $sl = 1$.
- **Report:** the barriers, the two labels, the derivation and the number in (c)
- **Model:** A3

Interpretation

- Formula (b) has no vertical barrier: which paths does a vertical barrier at h censor, and what benchmark would you compare with B4 for $(pt, sl) = (2, 1)$?

A5 [Solved]: The Variance of the Sharpe Ratio, PSR and DSR

Task

- **Context:** i.i.d. returns with mean μ , variance σ^2 , skewness γ_3 and kurtosis γ_4 ; a strategy with $\widehat{SR}_{ann} = 1.2$ on $T = 756$ daily returns, $\hat{\gamma}_3 = -0.4$, $\hat{\gamma}_4 = 5$ (Bailey & López de Prado (2012); Bailey & López de Prado (2014))
- **Tasks:**
 - ▶ (a) Derive $\text{Var}(\widehat{SR}) \approx \frac{1}{T} \left(1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2 \right)$ by the delta method (Lo (2002)).
 - ▶ (b) Compute $PSR(0) = \Phi \left(\widehat{SR} \sqrt{T-1} / \sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2} \right)$ with the daily SR.
 - ▶ (c) Compute SR_0 and $DSR = PSR(SR_0)$ if the strategy is the best of $N = 100$ independent trials, $V = 1/T$.
- **Report:** the variance formula, $PSR(0)$, SR_0 (daily and annualised) and the DSR

Interpretation

- Would you allocate capital to this strategy? What number of trials N would make you change your mind?

A5: Solution (1/2)

Solution

- (a) $SR = g(m_1, m_2) = m_1 / \sqrt{m_2 - m_1^2}$ with $m_j = \mathbb{E}[r^j]$; gradient $\nabla g = (\frac{1}{\sigma} + \frac{\mu^2}{\sigma^3}, -\frac{\mu}{2\sigma^3})$
- Moments of (r, r^2) : $\text{Var}(r) = \sigma^2$, $\text{Cov}(r, r^2) = \sigma^3\gamma_3 + 2\mu\sigma^2$, $\text{Var}(r^2) = \sigma^4(\gamma_4 - 1) + 4\mu\sigma^3\gamma_3 + 4\mu^2\sigma^2$
- $T \text{Var}(\widehat{SR}) \approx \nabla g^\top \Sigma \nabla g$; the terms in μ^2/σ^2 cancel and leave $1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2$ (Normal distribution: $1 + \frac{1}{2} SR^2$)

A5: Solution (2/2)

Solution

- (b) Daily $\widehat{SR} = 1.2/\sqrt{252} = 0.07559$; denominator $\sqrt{1 + 0.4 \cdot 0.07559 + 1 \cdot 0.07559^2} = 1.01782$
 - ▶ $z = 0.07559 \cdot \sqrt{755}/1.01782 = 2.041$, $PSR(0) = \Phi(2.041) = 0.979$
- (c) $\sqrt{V} = 1/\sqrt{756} = 0.03637$; $\Phi^{-1}(0.99) = 2.326$, $\Phi^{-1}(1 - \frac{1}{100e}) = 2.678$
 - ▶ $SR_0 = 0.03637 [0.4228 \cdot 2.326 + 0.5772 \cdot 2.678] = 0.0920$ per day (1.46 annualised)
 - ▶ $z = (0.07559 - 0.0920)\sqrt{755}/1.01782 = -0.444$, $DSR = \Phi(-0.444) = 0.329$
- Decision: $DSR < 0.95$; luck alone gives a best SR of 1.46 > 1.2 among 100 trials: do not allocate

A6 [Proposed]: DSR on a Shorter Sample and the Gumbel Constant

Task

- **Context:** a strategy with $\widehat{SR}_{ann} = 1.5$ on 2 years of daily data ($T = 504$), $\hat{\gamma}_3 = -0.2$, $\hat{\gamma}_4 = 6$; $N = 20$ independent trials; $Z_1, \dots, Z_N$ i.i.d. $N(0, 1)$
- **Tasks:**
 - ▶ (a) Compute $PSR(0)$, SR_0 and the DSR with $V = 1/T$, step by step as in A5.
 - ▶ (b) Show that $\mathbb{E}[\max_{n \leq N} Z_n] \approx (1 - \gamma)\Phi^{-1}(1 - \frac{1}{N}) + \gamma\Phi^{-1}(1 - \frac{1}{Ne})$ in the Gumbel limit, with location $b_N = \Phi^{-1}(1 - \frac{1}{N})$ and scale $a_N = \Phi^{-1}(1 - \frac{1}{Ne}) - b_N$.
 - ▶ (c) Check (b) by Monte Carlo for $N = 100$ (20,000 draws).
- **Report:** $PSR(0)$, SR_0 , DSR, the derivation and the Monte Carlo mean
- **Model:** A5

Interpretation

- Compared with A5 the Sharpe ratio is higher and there are fewer trials: is the evidence stronger? What drives the difference?

A7 [Solved]: How Much Does Purging Cost?

Task

- **Context:** n observations in K contiguous folds of length $L = n/K$; the label of observation i ends at $t_1(i) = i + h$; embargo of e observations after each test fold (López de Prado (2018))
- **Tasks:**
 - ▶ (a) For an interior test fold, count the training observations removed before and after the fold, and the share of non-test data lost.
 - ▶ (b) Apply the result to $n = 20$, $K = 5$, $h = 3$, $e = 1$ and the test fold $\{8, \dots, 11\}$; list the purged, embargoed and training observations.
 - ▶ (c) Explain how the choice of K trades bias against variance of the CV estimate.
- **Report:** the general formula, the three sets in (b) and the trade-off in (c)

Interpretation

- Why does shuffled K-fold overstate accuracy in this setting (lecture: 69.4% on pure noise)?

A7: Solution

Solution

- (a) Before the fold: observations with $i + h \geq$ first test index overlap it: h of them
- After the fold: observations starting before the last test label ends ($i \leq$ last test index $+h$) and at least the e embargoed ones: $\max(h, e)$
- Lost share of the $n - L$ non-test observations: $\frac{h + \max(h, e)}{n - L} = \frac{K(h + \max(h, e))}{n(K - 1)}$
- (b) Purged before: $\{5, 6, 7\}$ ($t_1 \geq 8$); removed after: $\{12, 13, 14\}$ (12 = embargo, 13–14 start before day 14, when the last test label is known)
 - training set $\{0, \dots, 4\} \cup \{15, \dots, 19\}$: 6 of 16 non-test observations lost (37.5%), as the formula says: $\frac{3+3}{16}$
- (c) Larger K : more training data per fit (less pessimistic bias), but shorter and noisier test folds and more purged boundaries; when h is large relative to L , purging dominates

A8 [Proposed]: Purging with Shorter Labels

Task

- **Context:** as in A7, with $n = 20$, $K = 5$ unshuffled folds, label end $t_1(i) = \min(i + 2, 19)$ and an embargo of 1 observation
- **Tasks:**
 - ▶ (a) For the test fold $\{12, \dots, 15\}$, list the purged observations, the embargoed ones and the training set.
 - ▶ (b) Check your count against the formula of A7.
 - ▶ (c) Use the formula to compute the share of non-test data lost for $h = 20$ and folds of $L = 50$ days ($K = 5$).
- **Report:** the three sets and the two shares
- **Model:** A7

Interpretation

- What does purging cost when labels are long and folds are short, and how would you respond?

A9 [Solved]: AUC Step by Step, with a Standard Error

Task

- **Context:** classifier scores of 3 positives (0.80, 0.55, 0.40) and 3 negatives (0.60, 0.35, 0.20); Hanley–McNeil standard error (Hanley & McNeil (1982))
- **Tasks:**
 - ▶ (a) Compute the AUC as the share of (positive, negative) pairs ranked correctly, $U/(n_1 n_0)$.
 - ▶ (b) Compute $SE^2 = \frac{A(1-A) + (n_1 - 1)(Q_1 - A^2) + (n_0 - 1)(Q_2 - A^2)}{n_1 n_0}$ with $Q_1 = \frac{A}{2-A}$, $Q_2 = \frac{2A^2}{1+A}$.
 - ▶ (c) Test $H_0: AUC = 0.5$ at 5% with $z = (A - 0.5)/SE$.
- **Report:** U , AUC, SE, z and the decision

Interpretation

- Which assumption of the SE formula fails for overlapping 5-day labels, and in which direction does it bias the test?

A9: Solution

Solution

- (a) 0.80 beats 3 negatives, 0.55 beats 2, 0.40 beats 2: $U = 7$ of 9 pairs, $A = 0.778$
- (b) $Q_1 = 0.778/1.222 = 0.636$, $Q_2 = 2 \cdot 0.605/1.778 = 0.681$
 - ▶ $SE^2 = [0.173 + 2(0.636 - 0.605) + 2(0.681 - 0.605)]/9 = 0.0429$, $SE = 0.207$
- (c) $z = (0.778 - 0.5)/0.207 = 1.34$, $p = 0.18$ (asymptotic Wald approximation, unreliable here); exact permutation test with exchangeable labels under H_0 : $P(U \geq 7) = 4/20$, two-sided $p = 0.40$; do not reject: six observations cannot separate skill from luck
- The formula assumes independent observations; overlapping labels share returns, so the effective sample is smaller and the formula understates the SE: the test over-rejects. B7 replaces it by a block bootstrap

A10 [Proposed]: AUC as a U-Statistic and the DeLong Variance

Task

- **Context:** scores of 4 positives (0.70, 0.62, 0.48, 0.45) and 4 negatives (0.66, 0.50, 0.40, 0.30); kernel $\psi(s_i, s_j) = 1\{s_i > s_j\} + \frac{1}{2}1\{s_i = s_j\}$
- **Tasks:**
 - ▶ (a) Compute U , the AUC and its Hanley–McNeil SE, as in A9.
 - ▶ (b) Show that $\text{AUC} = \frac{1}{n_1 n_0} \sum_{i,j} \psi(s_i, s_j)$ is a two-sample U-statistic; define the placement values $V_{10}(i) = \frac{1}{n_0} \sum_j \psi(s_i, s_j)$ and $V_{01}(j) = \frac{1}{n_1} \sum_i \psi(s_i, s_j)$.
 - ▶ (c) Derive $\widehat{\text{Var}}(\text{AUC}) = S_{10}/n_1 + S_{01}/n_0$ (DeLong, DeLong & Clarke-Pearson (1988)), where S_{10} , S_{01} are the sample variances of the placement values, and compute the DeLong SE.
- **Report:** U , AUC, the two standard errors and the placement values
- **Model:** A9

Interpretation

- Why do both formulas assume independent observations, and what does that mean for overlapping 5-day labels?

B1 [Solved]: Memory of the Bitcoin Log Price: ADF Rule vs. Local Whittle

Task

- **Context:** daily BTC-USD closes 2014-09-17 to 2026-09-18 (4,385 obs., `data/market/BTC-USD.CC.csv`), $x_t = \ln P_t$; FFD with $\tau = 10^{-4}$; $m = n^{0.65}$ Fourier frequencies
- **Tasks:**
 - ▶ (a) For $d \in \{0, 0.05, \dots, 1\}$ compute the FFD series, the ADF statistic (constant, 1 lag, [Dickey & Fuller \(1979\)](#)) and $\text{Corr}(\tilde{x}_t, x_t)$; find the smallest d^* for which the ADF statistic is below the 5% critical value.
 - ▶ (b) Estimate the memory with 95% CIs: exact local Whittle ([Shimotsu & Phillips \(2005\)](#)) for x_t and the FFD series at d^* , local Whittle ([Robinson \(1995\)](#)) for returns and absolute returns.
 - ▶ (c) Test $H_0 : d = 1$ for the log price with $z = (\hat{d} - 1) 2\sqrt{m}$.
- **Report:** the ADF table for $d \leq 0.25$, d^* , the four estimates with CIs and the test in (c)

Interpretation

- Is the FFD series at d^* stationary? Where does the memory of Bitcoin actually sit?

B1: Solution (1/2)

Solution

- (a) ADF statistic along d (5% critical value -2.86):

d	0	0.05	0.10	0.15	0.20	0.25
ADF	-1.21	-2.32	-2.24	-2.39	-2.85	-3.38
Corr	1.000	0.999	0.995	0.985	0.972	0.955

- The rule gives $d^* = 0.25$ with correlation 0.955; the path is not monotone in d (-2.32 , then -2.24), a first sign of fragility
- The FFD weights at $d = 0.25$ sum to 0.178, so the FFD series still contains $0.178 x_t$ (A1)

B1: Solution (2/2)

Solution

- (b) Memory estimates, $m = n^{0.65}$, 95% CI $\hat{d} \pm 1.96/(2\sqrt{m})$:
 - ▶ log price (ELW): 1.065 [1.000; 1.129]; FFD series at $d^* = 0.25$ (ELW): 0.814 [0.747; 0.880]
 - ▶ returns (LW): 0.063 [-0.002; 0.127]; absolute returns (LW): 0.353 [0.289; 0.417]
- (c) $z = (1.065 - 1)/0.033 = 1.97$, two-sided $p \approx 0.05$: at most a borderline rejection, in the direction of $d > 1$ (bubble episodes), never towards stationarity
- Reading: over the band the FFD series has $\hat{d} \approx 1 - 0.25$ (its integration order is still 1, A1) and its CI lies far above $\frac{1}{2}$: it is not stationary; the long memory of Bitcoin is in volatility ($|r_t|$), as for the S&P 500 in the lecture

B2 [Proposed]: Memory of Ether: a Trap

Task

- ▣ **Context:** daily ETH-USD closes 2015-08-07 to 2026-09-18 (4,061 obs., `data/market/ETH-USD.CC.csv`); the functions of B1
- ▣ **Tasks:**
 - ▶ (a) Repeat the ADF rule of B1 on $\ln P_t^{ETH}$ and report d^* together with the ADF statistics at $d = 0$ and $d = 0.05$.
 - ▶ (b) Redo the $d = 0$ test with a constant and a trend, and redo the rule on the sample starting on 1 January 2018.
 - ▶ (c) Estimate d by exact local Whittle (95% CI) on both samples, and by local Whittle for absolute returns.
- ▣ **Report:** the three ADF-rule answers and the three memory estimates with CIs
- ▣ **Model:** B1

Interpretation

- ▣ Which d would you use in a model, and why? What does this say about unit-root tests on an asset that grew 1,000-fold?

B3 [Solved]: Triple-Barrier Labels on Bitcoin

Task

- **Context:** daily BTC-USD closes 2014–2026; EWMA volatility of daily log returns (span 50); every day is an event; vertical barrier $h = 10$ days (López de Prado (2018))
- **Tasks:**
 - ▶ (a) Label every day with symmetric barriers $pt = sl = 1$ and $pt = sl = 2$ (units of $\sigma_{i_0} \sqrt{h}$).
 - ▶ (b) Compute the shares of upper, lower and vertical exits, the average holding period and the share of +1 labels.
 - ▶ (c) Compare with the 10-day fixed-horizon label.
- **Report:** a table by setting and a histogram of the holding periods

Interpretation

- Why are about 56% of the labels +1, and what is then the benchmark accuracy of a classifier?

B3: Solution

Solution

- (a)–(b) Exit shares and labels:

	Upper	Lower	Vertical	Days	+1
$pt = sl = 1$	22.8%	18.2%	59.0%	8.25	55.7%
$pt = sl = 2$	7.2%	4.9%	87.9%	9.58	55.6%
fixed 10 days	–	–	100%	10	55.5%

- (c) Wider barriers push most exits to the vertical barrier: the labels converge to fixed-horizon labels
- Reading: BTC rose over the sample, so +1 dominates whatever the barriers; “always up” (about 56%) is the benchmark to beat, and the 59.0% time-outs are above the 52.8% of A3 (daily monitoring); the rest may reflect fat tails, volatility clustering and volatility-estimation error, which this comparison does not separate

B4 [Proposed]: Asymmetric Barriers on Bitcoin

Task

- ▣ **Context:** the data, volatility and labelling function of B3, $h = 10$
- ▣ **Tasks:**
 - ▶ (a) Repeat B3 with $(pt, sl) = (2, 1)$ and $(1, 2)$.
 - ▶ (b) Report the exit shares, the share of +1 labels and the average holding period.
 - ▶ (c) Compare the share of upper exits among horizontal exits with the infinite-horizon value $\frac{sl}{pt+sl}$ of A3 and with a simulated finite-horizon benchmark (Gaussian daily steps, same barriers, vertical barrier at h).
- ▣ **Report:** a table with the two settings and the comparison in (c)
- ▣ **Model:** B3

Interpretation

- ▣ Which setting would a trend follower choose? How does the choice change the class balance a classifier sees?

B5 [Solved]: Purged K-Fold vs. Shuffled K-Fold

Task

- **Context:** a simulated leakage dataset (seed 42): random walk of 3,000 days, label = sign of the 20-day forward return, 5 features = 20-day means of independent noise; true accuracy 50%
- **Tasks:**
 - ▶ (a) Write `PurgedKfold(n_splits, t1, pct_embargo)` with a scikit-learn `split(X)` method.
 - ▶ (b) Fit a random forest ([Breiman \(2001\)](#); 200 trees, minimum leaf 5) and compute the out-of-fold accuracy under shuffled 5-fold, unshuffled 5-fold and purged 5-fold with a 1% embargo.
 - ▶ (c) Check your class on the A7 example.
- **Report:** the three mean accuracies with the five fold values

Interpretation

- Would the gap shrink with non-persistent features? With 1-day labels?

B5: Solution

Solution

- (b) Shuffled K-fold: **69.4%**; unshuffled: 50.0%; purged + 1% embargo: **49.7%**; truth: 50%
- (c) On the A7 example the class removes $\{5, 6, 7, 12, 13, 14\}$ for the test fold $\{8, \dots, 11\}$, as derived step by step
- Reading: shuffling places near-identical neighbours in train and test; a flexible model memorises them and “predicts” pure noise 19 points above chance
- Both ingredients are needed: overlapping labels *and* persistent features; remove either one and the leak disappears (Bergmeir, Hyndman & Koo (2018))

B6 [Proposed]: Find the Leak

Task

- **Context:** the B7 modelling frame (BTC, 5-day direction, 11 features, purged 5-fold); the notebook adds a 12th feature `ma11`, and the random forest now reaches a spectacular accuracy
- **Tasks:**
 - ▶ (a) Read how `ma11` is built and rank all features by MDI.
 - ▶ (b) Compute the correlation of `ma11` and of a trailing version with the target return.
 - ▶ (c) Rerun the purged CV with the trailing version and with the 11 base features only.
- **Report:** accuracy and AUC in the three cases, the MDI ranking and the two correlations
- **Model:** B5

Interpretation

- Why can purging not catch this kind of leak? Give a second example of look-ahead in a feature pipeline.

B7 [Solved]: Model Comparison and a CI for the AUC

Task

- **Context:** BTC 2014–2026; target $y_t = 1\{\ln P_{t+5} - \ln P_t > 0\}$; 11 features (momentum 1–120 days, volatility 20/60 and their ratio, distances to the 50- and 200-day moving averages, RSI); purged 5-fold, 1% embargo; 4,181 obs., 54.8% “up”
- **Tasks:**
 - ▶ (a) Fit Logit-L1 ([Tibshirani \(1996\)](#)), a random forest and gradient boosting ([Friedman \(2001\)](#)); report fold-mean accuracy and ROC AUC against “always up”.
 - ▶ (b) Compare MDI, MDA and mean |SHAP| ([Lundberg & Lee \(2017\)](#)) by Spearman rank correlation.
 - ▶ (c) For the pooled out-of-fold RF probabilities compute a 95% CI for the AUC with the DeLong SE ([DeLong, DeLong & Clarke-Pearson \(1988\)](#)) and with a circular block bootstrap ([Künsch \(1989\)](#)), $L \in \{1, 20, 60\}$, 1,000 replications.
- **Report:** the model table, the rank correlations and all four intervals

Interpretation

- Why is the bootstrap interval wider? Which block length would you defend?

B7: Solution (1/2)

Solution

- (a) Fold means:

	Logit-L1	RF	GB
Accuracy	54.8%	54.2%	52.2%
ROC AUC	0.492	0.516	0.502

- No model beats “always up” (54.8%); Logit-L1 predicts “up” on 99.6% of days
- (b) Spearman: MDI-SHAP 0.75, MDI-MDA -0.42, MDA-SHAP -0.10: in-sample and out-of-sample importances disagree

B7: Solution (2/2)

Solution

- (c) RF pooled AUC = 0.512
 - ▶ DeLong: SE 0.009, CI [0.495; 0.530], $z = 1.37$, $p = 0.17$
 - ▶ bootstrap $L = 1$: SE 0.009, CI [0.494; 0.530] (same as DeLong)
 - ▶ $L = 20$: SE 0.015, CI [0.484; 0.543], 21% of replications ≤ 0.5 ; $L = 60$: SE 0.014, CI [0.485; 0.541]
- Reading: DeLong assumes independent observations; overlapping 5-day labels and volatility clusters violate it, and the dependence-robust SE is 1.7 times larger
- Block length: at least the label horizon (5) and long enough for volatility clusters; results are stable from 20 to 60 days, so $L = 20$ is defensible

B8 [Proposed]: Diebold–Mariano on Probability Forecasts

Task

- **Context:** the pooled out-of-fold probabilities of B7 and a constant forecast equal to the training share of “up” days in each fold; log loss $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$
- **Tasks:**
 - ▶ (a) Compute the mean log loss of the four forecasts.
 - ▶ (b) For the pairs RF vs. Logit-L1, RF vs. constant and GB vs. constant compute $d_t = \ell_t^A - \ell_t^B$ and the DM statistic $\bar{d}/SE_{HAC}(\bar{d})$ (Diebold & Mariano (1995)) with Newey–West lags 4 and 20.
 - ▶ (c) Compare with the naive i.i.d. t -statistic and decide at 5%.
- **Report:** a table of log losses and of DM statistics with p -values
- **Model:** B7

Interpretation

- Can a model with AUC above 0.5 lose to a constant in log loss? What would change the conclusion?

B9 [Solved]: Meta-Labelling

Task

- **Context:** BTC 2014–2026; primary model: side of the MA20/MA50 crossover; meta-label = 1 if the primary trade is profitable at its triple-barrier exit ($pt = sl = 1$, $h = 10$) (López de Prado (2018))
- **Tasks:**
 - ▶ (a) Build the meta-labels for every day with a primary signal.
 - ▶ (b) Fit a random forest (balanced class weights) on the features plus the side, on the first 60% of the signals; test on the rest after a 10-day gap.
 - ▶ (c) Compare precision, recall, F1 and the average return per trade before (all trades) and after (only $\hat{p} > 0.5$).
- **Report:** a table with the two rows and the number of trades

Interpretation

- Precision up, F1 down: which metric matters for a trader who pays costs per trade?

B9: Solution

Solution

- Test period 2022-02 to 2026-09, 1,660 signals; the meta-model keeps 49.6% of them

	Prec.	Recall	F1	Trades	Ret./trade
Primary only	0.475	1.000	0.644	1,660	−0.27%
+ meta-model	0.517	0.541	0.529	824	+0.19%

- Precision rises by 4 points, recall halves and F1 falls; the average return per trade turns from −0.27% to +0.19%
- Reading: with costs per trade, precision and the return per trade matter, not F1; the next step is a test of that return (block bootstrap) and a DSR over all meta-model settings tried

B10 [Solved]: Grid Search, DSR and the Effective Number of Trials

Task

- **Context:** long/cash MA crossover on the BTC log price, 15 May 2015 – 2026 ($T = 4,145$ days, the first day on which every rule has a signal); fast window $\in \{5, 10, \dots, 55\}$, slow $\in \{20, 40, \dots, 240\}$, slow $>$ fast: $N = 120$ trials (Bailey & López de Prado (2014))
- **Tasks:**
 - ▶ (a) Pick the best trial by Sharpe ratio and compute its $\text{PSR}(0)$.
 - ▶ (b) From the 120×120 correlation matrix of trial returns compute the eigenvalues, the participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$, the smallest number of leading eigenvalues whose cumulative sum accounts for 95% of the total and $\hat{N} = \bar{\rho} + (1 - \bar{\rho})N$.
 - ▶ (c) Compute SR_0 and the DSR for $N = 120$, $\hat{N}$ and the 95% count, with $V = 1/T$.
- **Report:** the best rule, its SR and moments, the effective- N measures and a DSR table

Interpretation

- Which N and which $V[\widehat{SR}]$ would you report to an investment committee?

B10: Solution

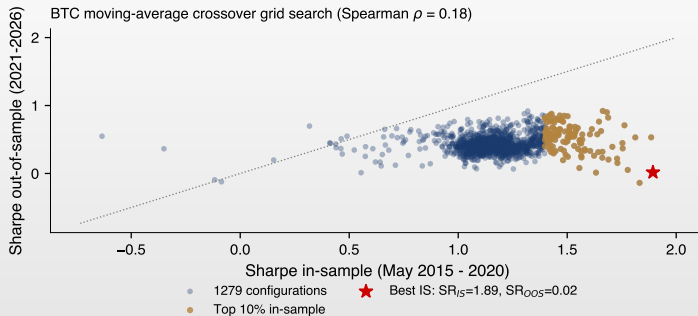
Solution

- (a) Best (5, 40): $SR_{ann} = 1.19$ (buy & hold 0.78); skewness 0.13, kurtosis 12.8; $PSR(0) = 1.000$
- (b) $\lambda_1 = 105.5$ explains 87.9% of the variance; participation ratio 1.3; 4 eigenvalues reach 95%; $\bar{\rho} = 0.875 \Rightarrow \hat{N} = 15.9$

DSR, $V = 1/T$	$N = 120$	$\hat{N} = 15.9$	$N = 4$
SR_0 (ann.)	0.77	0.53	0.31
DSR	0.922	0.987	0.998

- (c) Counting 120 correlated trials as independent over-deflates (0.922); with $\hat{N}$ the rule passes (0.987)
- Reading: the DSR tests “SR above luck”, not “above buy & hold”; report $\hat{N}$ with its method, and test against buy & hold separately (B13)

B10: Why Out-of-Sample Evidence Still Matters



- Lecture result: the best in-sample BTC crossover (SR 1.89, May 2015 – 2020) earned SR 0.02 in 2021–2026

B11 [Proposed]: Probability of Backtest Overfitting (CSCV)

Task

- **Context:** the $T \times 120$ matrix of daily returns of the B10 trials, split into $S = 16$ contiguous blocks (Bailey, Borwein, López de Prado & Zhu (2017))
- **Tasks:**
 - ▶ (a) For each of the $\binom{16}{8} = 12,870$ choices of 8 in-sample blocks, pick the trial with the best in-sample Sharpe ratio.
 - ▶ (b) Compute its relative out-of-sample rank ω and the logit $\lambda = \ln \frac{\omega}{1-\omega}$.
 - ▶ (c) Report $PBO = P(\lambda \leq 0)$, the median logit and the median in- and out-of-sample Sharpe ratio of the winner.
- **Report:** PBO, the histogram of λ and the Sharpe ratios in (c)
- **Model:** B10

Interpretation

- Is the grid overfit? How would S , or adding 1,000 random strategies to the grid, change the PBO?

B12 [Solved]: Pesaran–Timmermann and Clark–West on the BTC Direction Models

Task

- **Context:** the pooled out-of-fold probabilities of B7 ($n = 4,181$, 54.8% “up”), predicted sign $x_t = 1\{\hat{p}_t > 0.5\}$, and the constant forecast of B8 (Pesaran & Timmermann (1992); Clark & West (2007))
- **Tasks:**
 - ▶ (a) For each model compute $\hat{P}$, $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$ and the PT statistic.
 - ▶ (b) Because the 5-day labels overlap, compute the slope t -statistic of $y_t = a + b x_t + u_t$ with Newey–West errors (4 lags).
 - ▶ (c) Test Logit-L1 against the constant (nested) with the Clark–West statistic on the Brier loss, one-sided, HAC with 4 lags; compare with the plain DM test on the same loss.
- **Report:** a table with $\hat{P}$, $\hat{P}^*$, PT and the HAC t ; the CW and DM statistics with p -values

Interpretation

- Why can DM and Clark–West disagree for nested models? Which null does each test?

B12: Solution (1/2)

Solution

- (a)–(b) Directional tests:

	Pred. up	$\hat{P}$	$\hat{P}^*$	PT	t_{HAC}
Logit-L1	99.6%	0.548	0.548	0.89	0.82
RF	70.3%	0.542	0.519	3.20	2.21
GB	68.5%	0.522	0.518	0.65	0.45

- RF: its accuracy (54.2%) is below “always up” (54.8%), yet its signs are not independent of the outcome: $PT = 3.20$; the HAC correction shrinks the evidence to $t = 2.21$ ($p = 0.03$)
- Logit-L1 and GB: no significant directional information

B12: Solution (2/2)

Solution

- (c) Clark–West, Logit-L1 vs. constant: $t = -0.56$, one-sided $p = 0.71$: no evidence that the 11 features add information in the population model
- DM on the Brier loss: $t = 1.37$ (positive = logit worse), $p = 0.17$
- DM asks whether the *estimated* logit forecasts better; under the nested null its extra noise makes it lose, so DM rarely favours the larger model
- CW adds back $(\hat{\rho}_0 - \hat{\rho}_1)^2$ and asks whether the population coefficients are zero; here both agree: no significant signal
- CW is derived for nested linear models and genuine out-of-sample forecasts; on penalised-logit probabilities from pooled purged CV it is an approximate, descriptive adjustment
- Reading: a significant PT (RF) does not make a model useful: its log loss is still worse than the constant (B8)

B13 [Proposed]: SPA vs. DSR on the 120-Rule Grid

Task

- **Context:** the $T \times 120$ return matrix of B10 ($T = 4,145$); benchmarks cash ($r_0 = 0$) and buy-and-hold; stationary bootstrap (Politis & Romano (1994)) with mean block 20 days, 2,000 replications, seed 42
- **Tasks:**
 - ▶ (a) Compute the performance differentials $d_{k,t} = r_{k,t} - r_{0,t}$ for both benchmarks.
 - ▶ (b) Compute White's Reality Check p -value (White (2000)), Hansen's SPA p -value with the consistent recentring (Hansen (2005)) and the number of rules kept by the Romano–Wolf StepM at 5% (Romano & Wolf (2005)).
 - ▶ (c) Compare with the DSR of B10 under $N = 120$ and $\hat{N} = 15.9$.
- **Report:** a table with the largest studentised statistic, the two p -values and the StepM count per benchmark
- **Model:** B10 and B12; the lecture slide “SPA vs. DSR on the Bitcoin Grid”

Interpretation

- Which method handles correlated trials without choosing N ? Does the grid beat buy-and-hold?

B14 [Proposed]: Double Selection: Does the VIX Predict Bitcoin Returns?

Task

- **Context:** the B7 frame ($n = 4,181$); outcome $y_t = 5$ -day forward BTC log return in basis points; regressor $d_t = \log$ VIX close of the same calendar date, carried forward over weekends and US holidays (data/market/VIX.INDX.csv), standardised; 5 days = 5 BTC calendar days; controls: the 11 features, their squares and pairwise products ($p = 77$), standardised
- **Tasks:**
 - ▶ (a) Run a LASSO of y_t on (d_t, x_t) with the plug-in penalty $\lambda = 2c\sqrt{n}\Phi^{-1}(1 - \gamma/(2p))$, $c = 1.1$, $\gamma = 0.1/\log n$ (Belloni, Chernozhukov & Hansen (2014)); is the VIX kept?
 - ▶ (b) Run the two selection LASSOs (y_t on x_t , d_t on x_t) and report $|\hat{S}_y|$ and $|\hat{S}_d|$.
 - ▶ (c) Estimate OLS of y_t on d_t and the union of the selected controls, with HAC errors (4 lags); give $\hat{\theta}$, SE and the 95% CI.
- **Report:** the answer to (a), the two set sizes and the post-double-selection estimate
- **Model:** the lecture example “Does the VIX Predict S&P 500 Returns?” and B12

Interpretation

- Why does the single LASSO give no valid inference? What does the CI say about a crypto “fear premium”?

C1 [Proposed]: Does a Foundation Model Beat Boosting?

Task

- **Context:** the BTC 5-day direction of B7 (daily closes 2014–2026); a time-series foundation model of Chapter 14 used zero-shot; baselines of B7/B8; a team project with no single correct method
- **Tasks:**
 - ▶ (a) Choose a model and restrict the test window to dates after its pre-training cut-off, otherwise the test leaks.
 - ▶ (b) Convert the forecast quantiles of the 5-day return into $P(r_{t,t+5} > 0)$ by interpolation.
 - ▶ (c) Refit gradient boosting and the constant forecast at every forecast origin on a fixed-length rolling window, and compare the log losses on exactly the same dates with the Giacomini–White test ([Giacomini & White \(2006\)](#); HAC errors).
 - ▶ (d) Define a long/cash rule on $P(\text{up}) > 0.5$ and compute its DSR, or the SPA p -value, over all configurations you tried.
- **Report:** a Quantlet folder in the team repository (code, `Metainfo.txt`, one chart) and one slide with the conclusion and its main limitation

Interpretation

- If the foundation model wins, is the gain economically meaningful after costs, and would it survive the next five years of data?

C2 [Proposed]: Critique an AI Answer

- A student asks an AI assistant: “Predict tomorrow’s direction of Bitcoin from past returns and evaluate the model out of sample with 5-fold cross-validation”
- The assistant answers with this code (btc: daily closes, imports omitted):

```
r = np.log(btc).diff()
X = pd.DataFrame({'r1': r, 'r5': r.rolling(5).sum(), 'vol20': r.rolling(20).std(),
                  'trend10': r.rolling(10, center=True).mean()})
data = pd.concat([X, r.shift(-1).rename('fwd')], axis=1).dropna()
y = (data['fwd'] > 0).astype(int) # 1 if tomorrow is up
Xs = StandardScaler().fit_transform(data[X.columns])
cv = KFold(n_splits=5, shuffle=True, random_state=0)
model = LogisticRegression(penalty='l1', C=0.1, solver='liblinear')
auc = cross_val_score(model, Xs, y, cv=cv, scoring='roc_auc')
print(round(auc.mean(), 3))
```

- “Output: 0.697. An out-of-sample AUC of 0.70 on 4361 days is a strong, tradable signal; cross-validation is out of sample, so there is no overfitting”

C2 [Proposed]: Critique an AI Answer, Task

Task

- The answer contains **two** leaks of test information into training, a validation design that does not match deployment and **two** unsupported claims
- **Context:** the code and claim of the previous slide, run on daily BTC-USD closes 2014–2026 (data/market/BTC-USD.CC.csv)
- **Tasks**, for each error
 - ▶ (a) Say what is wrong and why.
 - ▶ (b) Say how you would detect it: a check in the code, a result from B5–B7, a rule of the lecture.
 - ▶ (c) Give the corrected code.
 - ▶ (d) Rerun the corrected code with a walk-forward split and report the AUC.
- **Report:** the five errors as in AI_ERRORS.md: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- Which days enter trend10 on day t ? Which of them is the target?
- Is an AUC of 0.70 for next-day Bitcoin direction plausible, given B6 and B7?
- Why is “cross-validation is out of sample” not enough for time series?

C3 [Proposed]: Data Specification

- Source: the value-weighted decile portfolios of the 15 univariate sorts of the [Kenneth French Data Library](#)
 - ▶ size, book-to-market, operating profitability, investment, earnings/price, cash flow/price, dividend yield, momentum, short-term reversal, long-term reversal, accruals, market beta, net share issues, variance, residual variance
- Excess returns: subtract the one-month bill rate of the library's factor file
- Legs: per characteristic a short leg (equal-weighted mean of deciles 1–5) and a long leg (mean of deciles 6–10), both entered with a positive sign (θ sets the sign): 30 factors $\tilde{F}_{t+1}$
- Sample: only months in which all 30 legs and the FF5 factors exist; no imputation
- Limitation: halves of value-weighted deciles replace the rank-weighted legs of 46 characteristics on individual stocks, so the published LS and EN values cannot be reproduced

C3 [Proposed]: A No-Arbitrage SDF on Characteristic-Sorted Deciles

Task

- ▣ Context: the lecture case study (Chen, Pelger & Zhu (2024)): FF3/FF5 (Fama & French (2015); Table A.VI) are reproducible with library data; the linear SDF (Section II.F) needs the stock panel, so here it is an extension, not a replication
- ▣ Tasks:
 - ▶ (a) Build the monthly panel, 1967 to the latest month: the 30 legs $\tilde{F}_{t+1}$ specified on the previous slide.
 - ▶ (b) On 1967–1986 compute $\hat{\theta}_{LS} = \hat{\Sigma}^{-1} \hat{\mu}$ (sample second moment and mean of $\tilde{F}_{t+1}$), $\hat{\theta}_{EN} = \arg \min_{\theta} \|\hat{\mu} - \hat{\Sigma} \theta\|_2^2 + \lambda_2 \|\theta\|_2^2 + \lambda_1 \|\theta\|_1$ (Section II.F), (λ_1, λ_2) chosen by the Sharpe ratio on 1987–1991 over a reported grid, and the FF3 and FF5 tangency weights.
 - ▶ (c) With weights fixed, report the monthly Sharpe ratio of each SDF portfolio $F_{t+1} = \hat{\theta}^\top \tilde{F}_{t+1}$ on 1992–2016, with a 95% CI (Lo (2002)).
 - ▶ (d) Estimate $\hat{\beta}_i \propto \overline{R_i^e F}$ (unconditional analogue of the paper's betas) on 1967–1986, project test returns on $\hat{\beta}$, compute EV and cross-sectional R^2 per characteristic (Section II.F); repeat (c)–(d) from 2017 on.
- ▣ Report: Sharpe ratios with CIs (Tables I, A.VI); EV and cross-sectional R^2 (qualitative comparison with EN, Table III); a Quantlet folder
- ▣ Model: A5 (Sharpe ratio inference) and B12 (predictive-ability tests)

Interpretation

- ▣ Why can the elastic net linear no-arbitrage SDF beat a forecasting network (Table I)? Does your post-2016 result survive the configurations tried?

Take-Home Assignment (Team Project)

- Choose one asset (equity index, stock, crypto, exchange rate or commodity) and deliver
 - ▶ features without look-ahead; for any fractional feature, a memory estimate (local Whittle) with its CI, chosen on training data only (B1–B2)
 - ▶ triple-barrier labels with a justified (pt, sl, h)
 - ▶ at least three models with Purged K-Fold, a Pesaran–Timmermann test, a DM or Clark–West test against a constant forecast and a block-bootstrap CI for the AUC
 - ▶ a log of every trial, the effective N , the DSR, the SPA p -value against buy-and-hold and the PBO of the final strategy
- GitHub repository with notebooks and Quantlet Metainfo.txt; a 10-minute presentation
- Log every AI error you catch in AI_ERRORS.md, as in C2: what was asked, what was wrong, how you found out, the correct version
- A negative result, honestly documented, is a good result

Further Study on Quantinar

- ▣  Random Forests — bagging, out-of-bag error, variable importance
- ▣  Shapley Values — model-agnostic local explanations
- ▣  Machine Learning in Financial Risk — ML estimation of risk measures
- ▣  Introduction to Reinforcement Learning — Q-learning, sequential decisions

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